UNDER THE HOOD

UNICORN RUN

How the game sees two players through one webcam, and how it draws a split-screen rainbow race



A 2-player body-controlled racer
built on Bun Rush · Bianco's Labs

THE GAME

Two players, one webcam
Side by side, 2.5 m back. The person on the left is P1, on the right P2: the same sides as their screen halves.

Your legs are the engine
Run in place to gallop. Stop, and your unicorn slows to a trot.

Lean, jump, hands up
Lean your upper body to steer, jump on the striped line, raise both hands to fire a rainbow dash.

Three laps of rainbow road
Rainbows fill the dash meter; rocks and logs make you tumble.



BUILT ON BUN RUSH

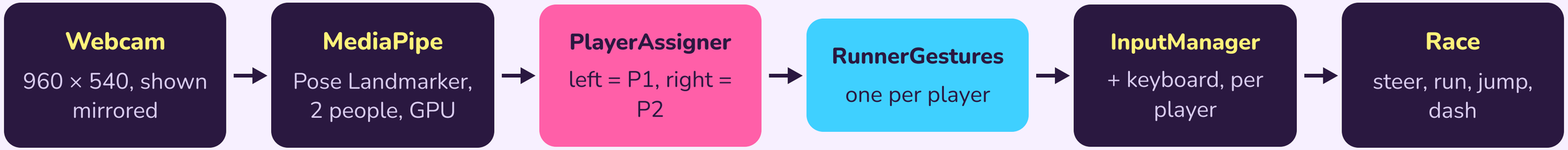
Layer	Kept from Bun Rush	New or changed for Unicorn Run
Game loop	Fixed 60 Hz simulation, draw per frame	Same loop, two players per tick
Body tracking	MediaPipe Pose Landmarker lite, vendored	numPoses 1 → 2, wider 960 × 540 picture
Who is who	Not needed (one player)	New PlayerAssigner: left = P1, right = P2
Gestures	Shoulder-width units, calibration, smoothing	Analogue steering, running in place, dash
Input	Keyboard + body merged, events latched	One intent and one key set per player
Graphics	Vector sprites, offscreen cache, HiDPI	Pseudo-3D road, split screen, baked sprites
Audio, server	Synthesised Web Audio, serve.ps1	Stereo: P1 left speaker, P2 right

PART 1

BODY DETECTION

How one webcam turns two people into two sets of race controls

FROM PIXELS TO INTENT



Stays on this computer

The picture is analysed inside the browser (WebAssembly + WebGL). No video is uploaded.

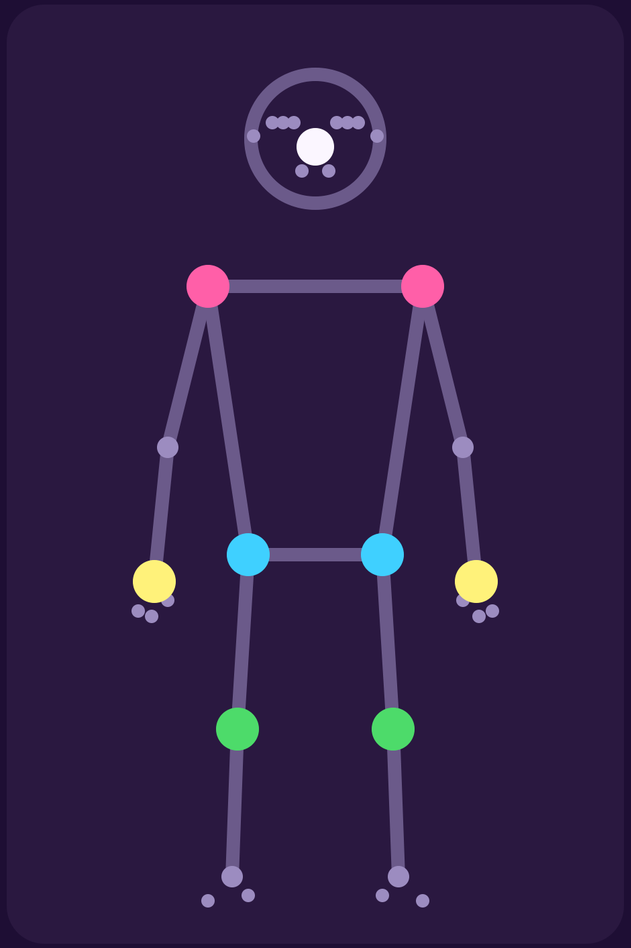
About 30 detections a second

Each camera frame is analysed once. The game still simulates at 60 Hz with the latest result.

No lost jumps

Jump and dash are events: held until the next game tick picks them up, exactly once.

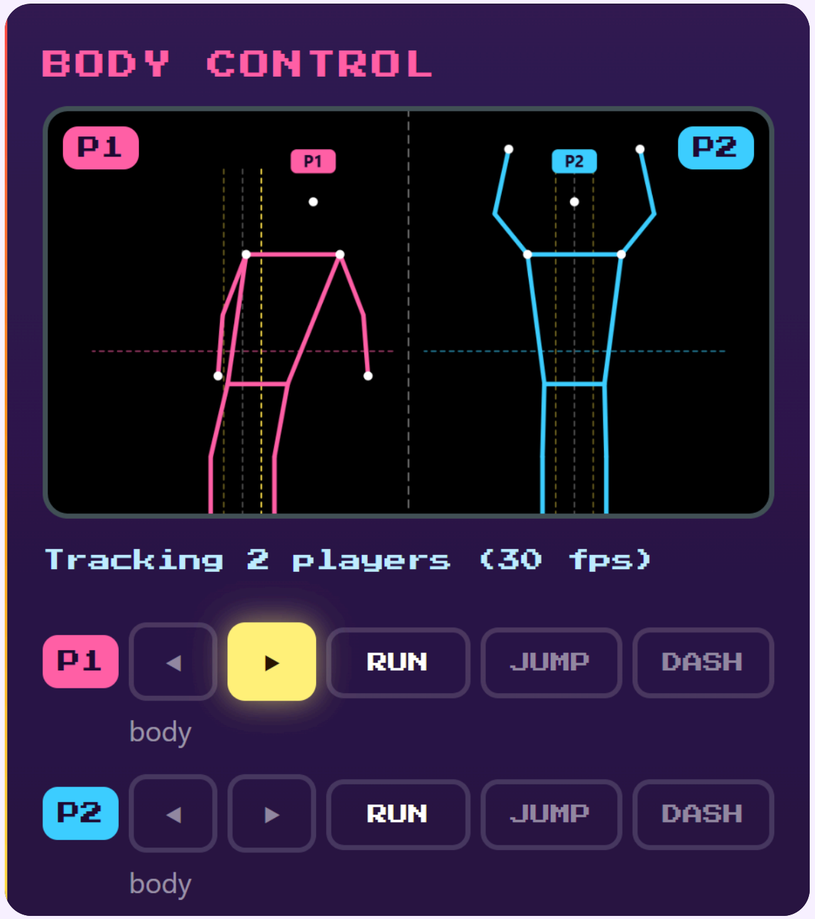
33 POINTS, 9 THAT MATTER



- **Shoulders** · their midpoint over the hips is your lean; their distance is the ruler for everything else.
- **Hips** · body height for the jump and the running bounce; the base of your lean.
- **Wrists** · hands up for join and dash; an X to skip; arm pumping.
- **Knees** · they alternate up and down when you run in place.
- **Nose** · the line both hands must pass for “hands up”.

Each point has x and y (0 to 1 across the picture) and a visibility score. Points the model is unsure about (under 0.5) are ignored.

WHO IS PLAYER 1?



Two skeletons

Sort them by shoulder centre in the mirrored picture. The left one is P1, the right one is P2: the same sides as their screen halves.

One skeleton

Give it to whoever was last seen within 0.22 picture widths of it (remembered for 2 s). P1 stays P1 when P2 steps out, even near the middle.

Then, one interpreter each

Each player has their own calibration. A lean is measured on the body itself, so it works wherever you stand on your half.

The one rule for players: stay on your half, shown by the dashed line in the camera panel.

MEASURED IN SHOULDER WIDTHS

```
lean = (hip centre x - shoulder centre x)
       ÷ shoulder width
```

Calibrate

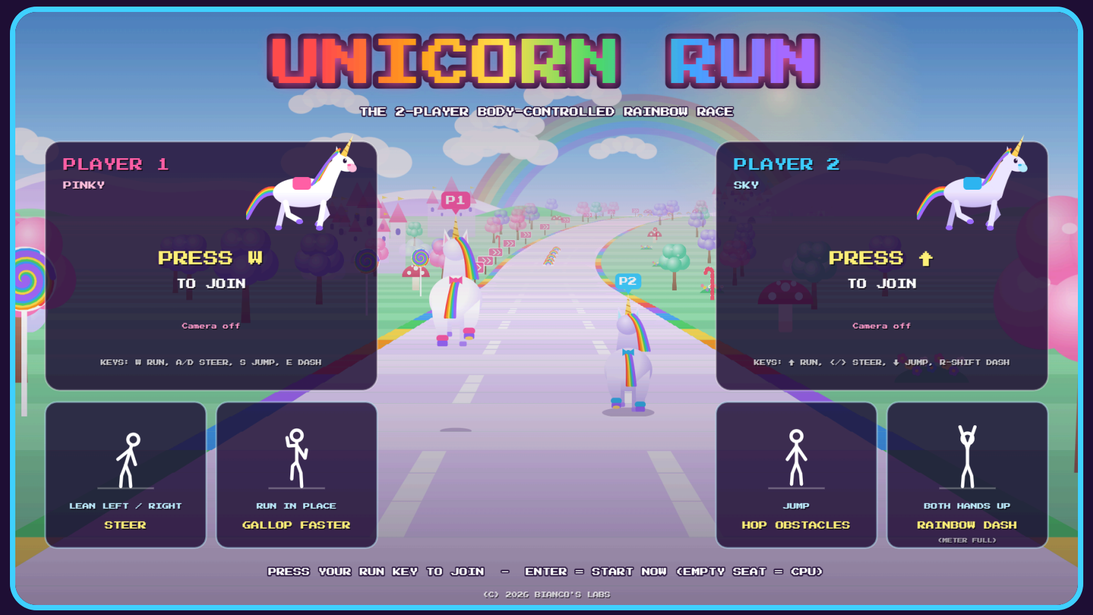
After joining, stand up straight with arms down for 1.5 s. The median of those frames becomes your ruler, your height and your “straight”.

Same rules at 2 m or 3 m

Every threshold is a fraction of your own shoulder width, so distance and body size cancel out.

Follow slowly

While you stand still, your normal height re-centres 2 % per frame; the ruler adapts 1 % if you step closer.

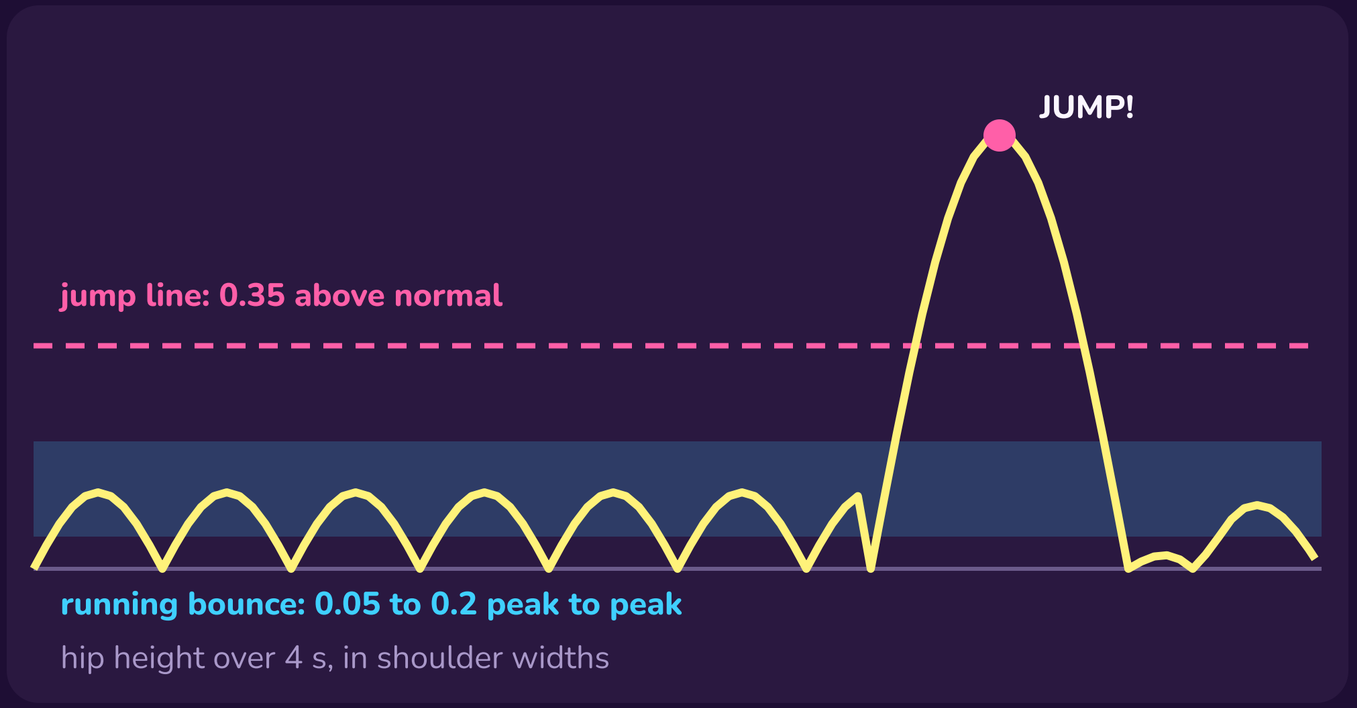


The lobby: raise both hands to join, then stand still. Ready players can test the steering slider and the run meter live.

FIVE MOVES, FIVE RULES

Move	What is measured	Rule (in shoulder widths)	In the game
Lean left / right	Shoulder centre over hip centre (shoulder slope if the hips are hidden)	Straight under 0.15; full steer at 0.55	Steer, −1 to +1
Run in place	Hip bounce, or knees or wrists alternating, over 0.6 s	Strongest of the three, peak to peak	Speed: 22 % trot up to 100 %
Jump	Hip height and upward speed	Above 0.35 AND rising faster than 1.4 per s	Jump, then 0.6 s cooldown
Both hands up	Both wrists above the nose	Held 1 s to join; 0.3 s to dash	Join, or rainbow dash
Arms crossed (X)	Wrists swapped sides, below the nose	Held 1 s	Skip the training

RUNNING VS JUMPING



Run meter: the bounce over the last 0.6 s, mapped from 0.05 (nothing) to 0.2 (flat out).

Jump: hips 0.35 above normal while rising faster than 1.4 per second. A running bounce never gets there.

No flicker: the meter rises fast and falls slowly, so it does not drop between strides.

KEEPING IT STEADY

Smoothing

Each point is averaged, 60 % new and 40 % old: less jitter for about one frame of delay.

Dead zones

Small sways do not steer, and walking or drifting across the room never does.

Visibility

Unsure shoulders mean “not seen”. Hips are only used when the camera sees them.

Arms are not jumps

With the hips out of view, raising your arms lifts the shoulders but never counts as a jump.

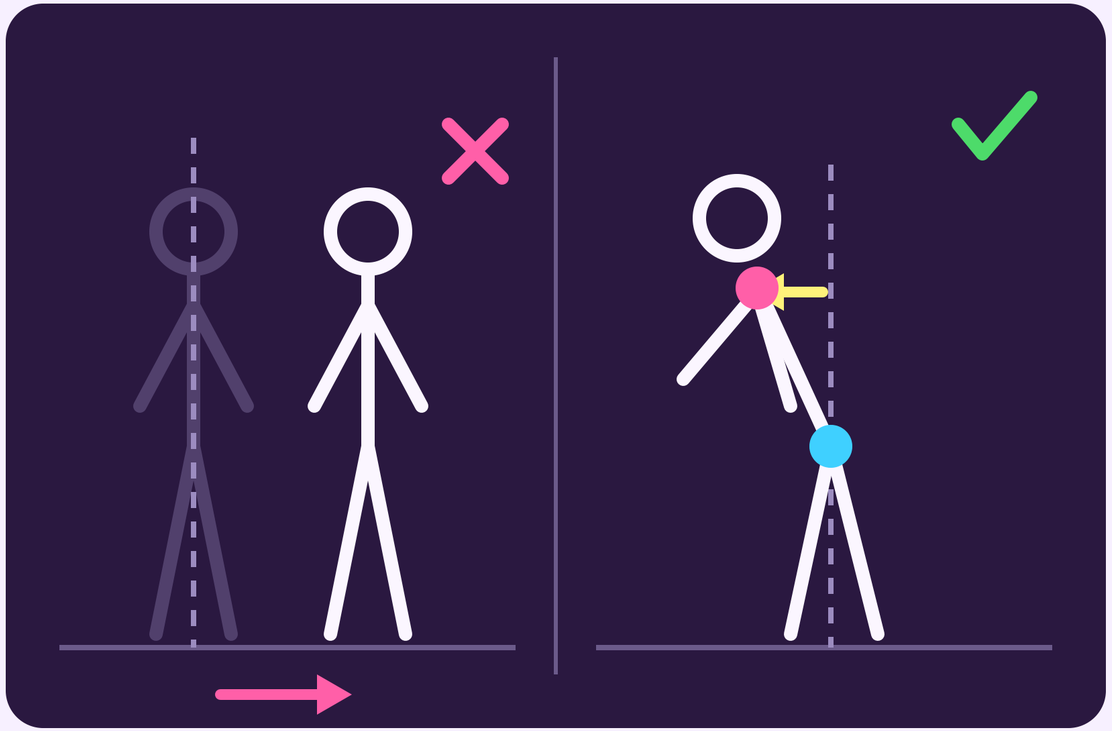
Auto-pause

A player out of view for 1.5 s pauses the race; 0.6 s back in view resumes it.

Tested without a camera

45 tests: synthetic skeletons, the jump line at every speed, the training, whole races.

LEAN, DON'T WALK



Before: shoulders vs. a spot in the room. Kids drift while running, so they steered without leaning.

Now: shoulders over hips. Only a real lean steers, anywhere on your half.

Measured on the body

Lean = hip centre minus shoulder centre, in shoulder widths. Hips out of view: the slope of the shoulder line instead.

Straight is a zone

Under 0.15 nothing happens; 0.55, about 20°, is a full turn. A lean while calibrating is trusted only up to 0.2.

See your lean

A slider at the bottom of each half: the dot is you, the grey middle is “straight”, the arrows light up when the unicorn turns.

TRAINING FIRST

1. **Lean left**
2. **Lean right**
3. **Run in place**
4. **Jump twice**
5. **Rainbow dash**



The race's own physics

On an empty straight road, in your own half: what you learn is exactly what the race does.

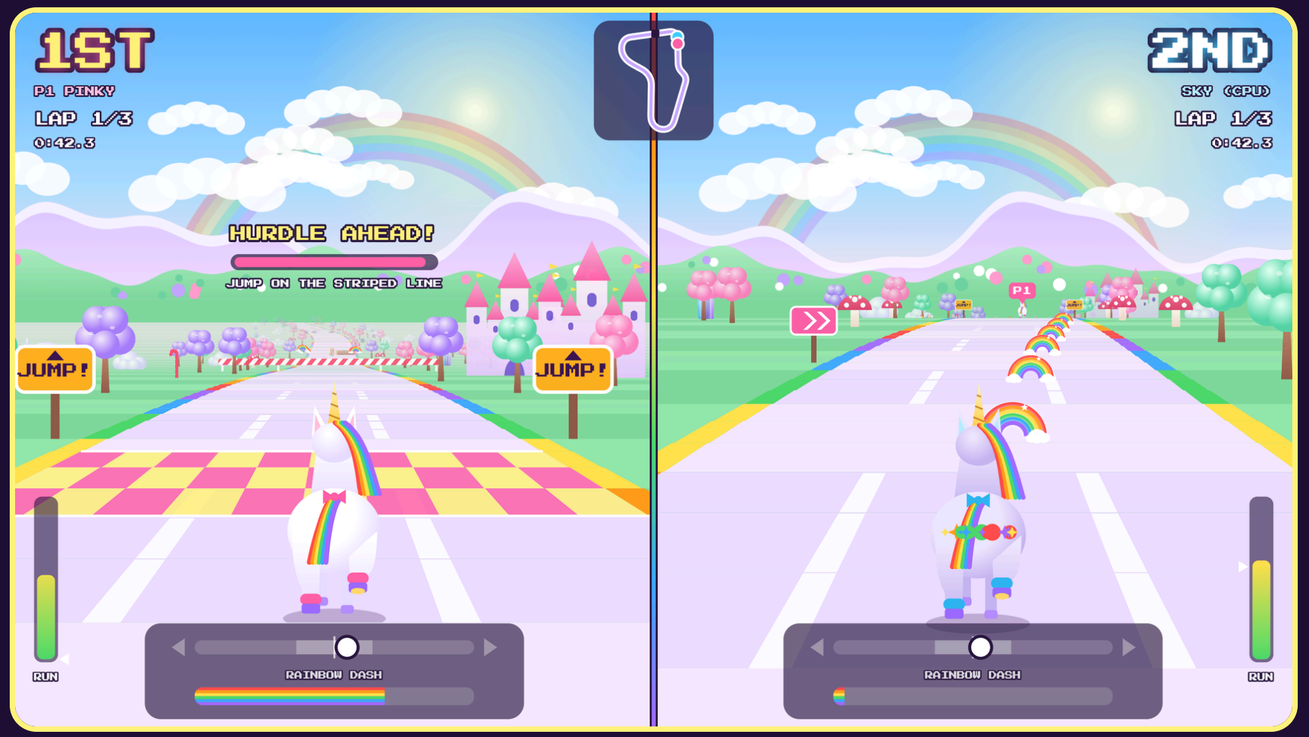
Tries add up

A step passes after enough time in total, so wobbly tries still count. Then GREAT! and sparkles.

Skip with an X

Cross your arms in front of you for 1 s. Rematches skip it; new players get it.

JUMP ON THE LINE



Warning: 2.5 s before a hurdle, HURDLE AHEAD! and a filling bar.

The line is painted where you take off: a time, not a distance, before the hurdle. 0.75 s for bodies, 0.45 s for keys.

Why time: any take-off 0.05 to 0.7 s before the hurdle clears it, at any speed. A body jump is seen about 0.3 s late.

Tested at trot, half and full speed, with the jump seen 0.15 to 0.45 s late.

PART 2

GRAPHICS

How a split-screen pseudo-3D race is drawn 60 times a second,
with no 3D engine and no image files

ONE LOOP, THREE STEPS

1

Read bodies

Only when the camera has a new frame: detect, assign to P1 / P2, update both gesture interpreters.

→

2

Simulate

Fixed steps of 1/60 s from an accumulator: the race runs at the same speed on any monitor.

→

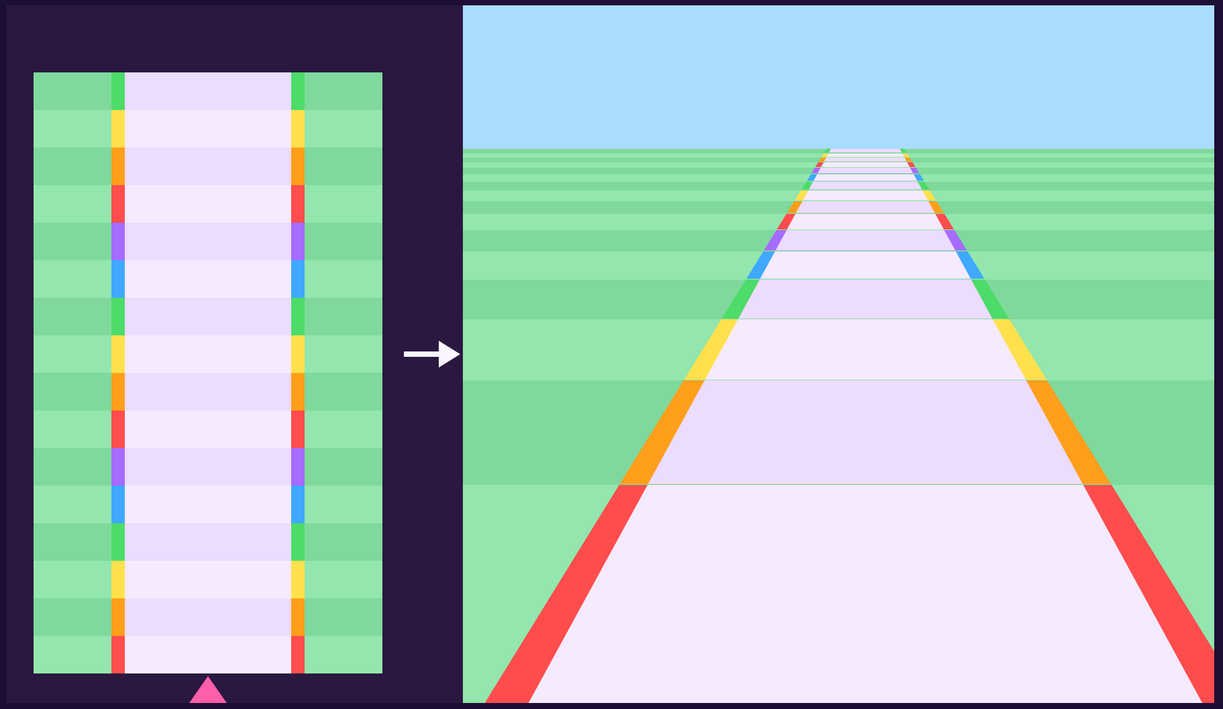
3

Draw

Once per screen refresh. The Renderer only reads the race; it never changes a rule.

1280 × 720 All drawing uses one fixed logical screen. A single transform scales it to the window and the screen’s pixel density, so it is sharp on any display.

A ROAD MADE OF SLICES



seen from above: equal slices

projected: trapezoids that shrink with distance

```
scale = depth ÷ z
x′ = centre + scale · x · k
y′ = horizon - scale · y · k
```

1,290 slices, each 200 units deep, each with a curve value and a height.

Curves: every slice is shifted sideways by an amount that keeps growing (dx += curve).

Hills: slices simply sit higher or lower.

PAINTED FAR TO NEAR

1. Sky, then 3 parallax layers: clouds, mountains, hills
2. Project 200 slices, near to far; mark the ones hidden behind a hill
3. Then far to near, for each slice: grass, rainbow kerbs, road, lane marks, fog
4. … the sprites standing on it: trees, signs, rainbows, rocks
5. … and any unicorn on it, yours included

Hills hide things for free

Nearer road is painted last, so it covers whatever is behind a crest. No depth buffer needed.

No seams

Each slice overlaps the previous one by one pixel, so no thin lines show between them.

TWO CAMERAS, ONE CANVAS



Two RoadViews, each clipped to its 640 × 720 half, each with its own camera.

The rival is just one more sprite on its slice, with a name tag.

HUD on the outer edges; the minimap in the middle is the curves added up into an outline.

Particles live in each half's screen space.

BAKED ONCE, DRAWN OFTEN



Baked

21 scenery painters drawn once at 2× size, then copied with drawImage at any scale.

Live

Unicorns are drawn every frame: gallop phase, lean, jump, tumble, dash glow.

Backgrounds

Sky and 3 tileable layers, pre-rendered at the real pixel size.

THE NUMBERS

5.5MS

to draw both views; the budget at 60 fps is 16.7 ms

230

things per view per frame: about 100 road slices and 130 sprites

45

automated tests, including whole simulated races

Drawing time measured in headless Chrome without a graphics card, averaged over 120 frames.

TRY IT

Open	What you get
`start.bat`	The game, camera and keyboard
`/?demo`	Two CPU unicorns race
`/tests/gallery.html`	Every sprite on one page
`npm test`	The 45 automated tests
`src/config.js`	Every threshold in one place



Stand side by side, raise both hands, and run.